

Here is the comprehensive, deep technical description of your adapted software architecture, **AgroSpray AI**.

This architectural blueprint maps your existing front-end layout 1:1 onto the strict technical requirements of the **EWIA.tech "Down to the Meter"** challenge , ensuring you fulfill the necessity test , eliminate black-box trust , and demonstrate an undeniable business case.

Core Software Concept: The Spatial Decision Engine

Instead of acting as a passive tracking map, the software serves as a **Real-Time Spatial Risk & Compliance Engine**. It streams a live drone position track, treats each GPS coordinate as a point wrapped in a dynamic error radius variable , and programmatically computes autonomous nozzle safety overrides using real OpenStreetMap geodata.

Deep Module-by-Module Specification

Module 01: Command Center (The Operational Cockpit)

This module acts as the live runtime monitor for the drone pilot or estate manager. It translates complex spatial buffer calculations into immediate, high-contrast operational directives.

- **The Action Banner:** Displays the current machine state.
 - *Example State:* "**CRITICAL BORDER APPROACH: Disabling Boom Nozzles 3 & 4 immediately to protect adjacent Organic Soy parcel.**"
- **The 5m vs 1m Master Toggle:** Placed prominently in the global header. Flipping this toggle alters the mathematical error radius parameter (\$r\$) in the backend database from a coarse 5 meters to a corrected 1 meter.
- **KPI Metrics Layout:**
 - **Live Precision (Formerly Current CV):** Displays the active hardware error margin (e.g., 0.85m via Galileo HAS vs 5.2m uncorrected).
 - **Field Boundary Clearance (Formerly Bunker Fill):** The exact distance in meters from the nearest restricted boundary polyline.
 - **Spray System Flux (Formerly Trucks Today):** Shows active boom nozzles and fluid pressure (e.g., 3/6 Nozzles Active).
 - **Risk Value Preservation (Formerly Power Price):** Displays the real-time financial liability avoided by using high precision (e.g., **€1,092 saved** by preventing off-target drift contamination fines).

Module 02: Field Digital Twin (The Geospatial Sandbox)

This module renders the physical environment using real vector geometries pulled from OpenStreetMap via Overpass API. It visualizes the core algorithmic problem: the overlap of position uncertainty.

- **The Geometry Engine:** The map renders two distinct polygons: the **Target Field Polygon** (authorized for spraying) and the **Restricted Adjacent Polygon** (forbidden crops, residential edges, or open water buffers).
- **Visualizing the "Flip" Scenario:**
 - *When Toggle = 5 Meters:* The UI draws a large, semi-transparent red circle with a 5-meter radius around the drone's estimated coordinate. Because this circle intersects the forbidden boundary line, the digital twin highlights the collision zone in flashing red.
 - *When Toggle = 1 Meter:* The circle shrinks tightly around the drone asset to a crisp 1-meter radius. The circle cleaves perfectly inside the safe target zone, turning the visualization green.

Module 03: AI Briefing (The Transparent Reasoner)

The evaluation criteria strictly prohibit unexplained machine output ("*No black-box trust*"). This module exposes the algorithmic verification pipeline to prove exactly *why* a choice was made.

- **The Timeline Forecast:** Simulates upcoming vector checkpoints along the pre-planned flight line (e.g., *00:44 - Approaching Hedgerow Buffer; 01:15 - Commencing 180-degree Row Turn*).
- **Reasoning Evidence Inputs:** Displays a hardware data-bus trace showing the live telemetry ingredients:
 - INPUT_01: Live NMEA GPS stream.
 - INPUT_02: OpenStreetMap boundary nodes ID #883921.
 - INPUT_03: geo_core.py computation array.
- **The Plain-Text Logic Box:** Explicitly prints the underlying geometric formula:

"Distance to forbidden polyline is 2.4m. Active error radius is 1.0m. Safe margin of 1.4m verified. Spray command authorized."

Module 04: Follow vs Ignore (The Business Case Matrix)

This module acts as the ultimate pitch Closer. It proves the commercial potential required by the jury by analyzing what happens if operations rely on uncorrected positioning.

- **Path A (Follow / 1-Meter Precision):** * *Predicted Outcome:* 100% of target crop treated right up to the legal fence line.
 - *Financials:* Zero chemical waste on bare soil, zero drift lawsuits, full compliance certification for automated flight profiles.
- **Path B (Ignore / 5-Meter Coarse Drift):**
 - *Predicted Outcome:* Chemical drift overreach or total system shutdown.
 - *Financials:* Forced 5-meter manual safety setbacks leave up to 25% of valuable field borders completely untreated, leading to weed/pest outbreaks and massive crop yield degradation.

Module 05: Fleet & Airspace (Formerly Trucks & Energy)

This module manages mission parameters, multi-drone logistics, and micro-climatic environmental layers that affect drone flight.

- **Incoming Flight Queue:** Lists autonomous flight tracks loaded into memory, tracking their battery levels, payload tank status, and active telemetry correction feeds.
- **Environmental Drift Strategy:** Displays live wind velocity and direction vectors. The software crosses this wind data with the position uncertainty radius to adjust the safety buffer dynamically: if wind speed rises, the software automatically inflates the error radius downwind to trigger earlier nozzle cut-offs.

Module 06: Demo Playback (The Evaluation Replay)

This module directly leverages the hackathon's evaluation kit files (`real_data.py` and `zurich_agz_groundtruth.csv`). It provides a repeatable tool to prove the software's resilience.

- **The Stress-Test Replay:** A timeline playback tool where judges can hit "Play" to watch a historical or simulated test flight sweep across a real farm map.
- **Graceful Degradation Demonstration:** As the track plays, you simulate a real-world signal drop (e.g., moving near heavy tree cover where accuracy drops from 1 meter to 3 meters). The UI updates dynamically, showing the error circle expanding on the fly, and demonstrates the software gracefully cutting off the outer spray booms to preserve safety boundaries instead of crashing or blindly spraying.